

Applying Power-SMC to Block Diffusion

1. Preliminaries and the Power Distribution

Let $x_{1:n} = (x_1, x_2, \dots, x_n)$ denote a sequence of length n . Under standard autoregressive generation, the joint probability of the sequence is $p(x_{1:n})$. In Power-SMC, we aim to sample from the sequence-level power distribution, where the unnormalized target distribution is defined as:

$$\pi_\alpha(x_{1:n}) = p(x_1, x_2, \dots, x_n)^\alpha \quad \text{for } \alpha > 1 \quad (1)$$

2. Block Generation and SMC Particle Weights

In a Block Diffusion framework, instead of generating one token at a time, we generate a block of B tokens simultaneously. Thus, an SMC particle at step m represents a sequence of length $n = m \times B$.

When extending the sequence from length n to $n + B$, the new unnormalized probability space requires summing over all possible combinations of the new block in the vocabulary V . The SMC weight update for the new block is determined by the ratio of the “new space” to the “old space”:

$$w_{n+1:n+B}^{\text{update}} = \frac{\sum_{x_{n+1}, \dots, x_{n+B} \in V} p(x_1, \dots, x_{n+B})^\alpha}{p(x_1, \dots, x_n)^\alpha} \quad (2)$$

For a block size of $B > 1$, the numerator involves an intractable sum over $|V|^B$ combinations. Furthermore, the tokens within the block are generated in parallel in a non-autoregressive manner.

3. Mean Field Approximation

To resolve the intractability, we apply a Mean Field Approximation. We assume that, given the fixed prefix $x_{1:n}$, the tokens in the newly generated block are conditionally independent of each other.

Under this assumption, the joint probability of the extended sequence can be factored as:

$$p(x_1, \dots, x_{n+B}) \approx p(x_{1:n}) \prod_{i=n+1}^{n+B} p(x_i \mid x_{1:n}) \quad (3)$$

4. Deriving the Tractable Weight Update

We substitute the Mean Field Approximation into the numerator of our weight update equation:

$$\sum_{x_{n+1}, \dots, x_{n+B}} p(x_1, \dots, x_{n+B})^\alpha \approx \sum_{x_{n+1}, \dots, x_{n+B}} \left(p(x_{1:n}) \prod_{i=n+1}^{n+B} p(x_i \mid x_{1:n}) \right)^\alpha \quad (4)$$

Distributing the exponent α over the product yields:

$$= \sum_{x_{n+1}, \dots, x_{n+B}} \left(p(x_{1:n})^\alpha \prod_{i=n+1}^{n+B} p(x_i | x_{1:n})^\alpha \right) \quad (5)$$

Since the prefix probability $p(x_{1:n})^\alpha$ is independent of the summation variables, we can factor it out. Furthermore, because the terms inside the product are independent, the sum of the products is equivalent to the product of the sums:

$$= p(x_{1:n})^\alpha \prod_{i=n+1}^{n+B} \left(\sum_{x_i \in V} p(x_i | x_{1:n})^\alpha \right) \quad (6)$$

Finally, substituting this simplified numerator back into the weight update ratio (Equation 2), the $p(x_{1:n})^\alpha$ terms perfectly cancel out:

$$w_{n+1:n+B}^{\text{update}} = \frac{p(x_{1:n})^\alpha \prod_{i=n+1}^{n+B} \left(\sum_{x_i \in V} p(x_i | x_{1:n})^\alpha \right)}{p(x_{1:n})^\alpha} \quad (7)$$

$$= \prod_{i=n+1}^{n+B} \left(\sum_{x_i \in V} p(x_i | x_{1:n})^\alpha \right) \quad (8)$$

Conclusion

Equation 8 demonstrates that under the Mean Field Approximation, the weight update for a block of size B reduces from an exponential $O(|V|^B)$ complexity to a linear $O(B \times |V|)$ complexity. This makes the application of Power-SMC to Block Diffusion computationally tractable.